

SECURING THE FUTURE OF BANKING: AI-Driven Correlation of Cybersecurity Telemetry & Transactional Behaviour

The Modern Threat Landscape and Multimodal Convergence

Modern financial institutions generate vast amounts of cybersecurity and transactional data, yet traditional defenses fail due to a lack of intelligent, contextual correlation. Sophisticated adversaries exploit the structural boundaries separating security operations from financial crime compliance. For example, credential theft via information-stealer malware, session hijacking, or automated databases query exploits can occur simultaneously with seemingly legitimate transactional instructions. Under unimodal systems, these events generate isolated, low-priority alerts that are easily overlooked.

Resolving this vulnerability requires the implementation of multimodal data fusion systems. This architecture integrates transactional, behavioral, and cybersecurity data into a unified, real-time analytics environment to deliver contextual threat awareness.

The current threat vectors targeting banks operate in multi-stage execution campaigns.

"Traffers" (traffic distributors) deploy information-stealer malware via Stealer-as-a-Service (MaaS) platforms to harvest multi-factor authentication (MFA) session cookies and credentials directly from browser caches. By stealing these long-lived authentication tokens, attackers can bypass security boundaries and access corporate accounts.

Once inside the session, malicious actors initiate a digital "smash and grab," executing rapid, automated data-harvesting scripts to locate sensitive files or initiate unauthorized fund transfers. If the security monitoring system evaluates the network logins independently of the payment ledger, the threat remains undetected until after exfiltration or settlement is complete.

Core Input Modality	Specific Telemetry & Signal Sources	Target Attack Vectors & Anomalies Detected	Underlying Correlation Logic
Cybersecurity Telemetry	Endpoint Detection & Response (EDR), SIEM logs, Active Directory query records, API access tokens	MFA session cookie theft, credential harvesting, unmanaged role drift, privilege escalation, unauthenticated API calls	Matches credential changes or temporary privilege elevations to immediate downstream payments.
Transactional Behaviour	Core banking ledgers, SWIFT/ACH payment corridors, invoice clearances	Suspicious beneficiary accounts, unfamiliar intermediary routing, rapid transaction velocity spikes, circular trade patterns	Detects abrupt deviation in standard corporate payment corridors or settlement values.
Behavioral Dynamics	Mouse movement,	Automated script	Differentiates normal

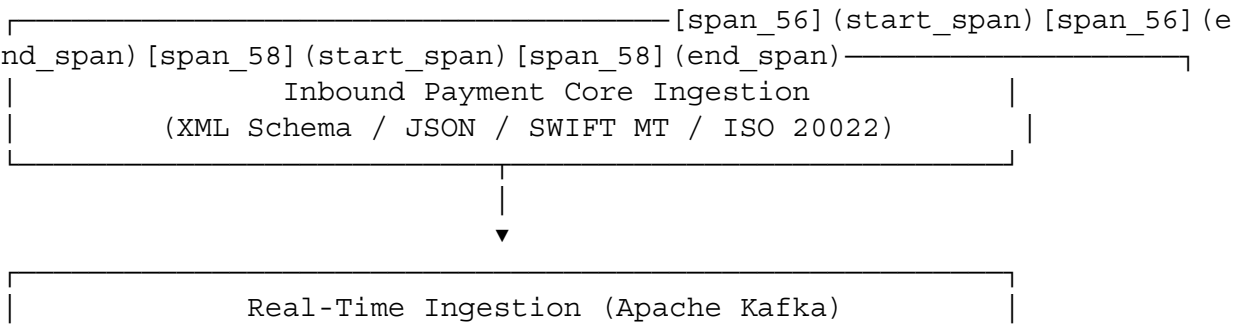
Core Input Modality	Specific Telemetry & Signal Sources	Target Attack Vectors & Anomalies Detected	Underlying Correlation Logic
	touch pressure, keystroke dynamics, mobile device interactions	playback, remote desktop access tools, cognitive input hesitation, robotic device orientation changes	human interaction from automated data-harvesting or remote access trojan signatures.
External & Mobile Metadata	Telecom cell tower records, FASTag mobility patterns, e-way bills, GST filings	Synthetic identity rings, vanishing trader schemes, impossible physical travel, geographic transaction clustering	Validates transactional billing addresses against physical device movement and state trade data.

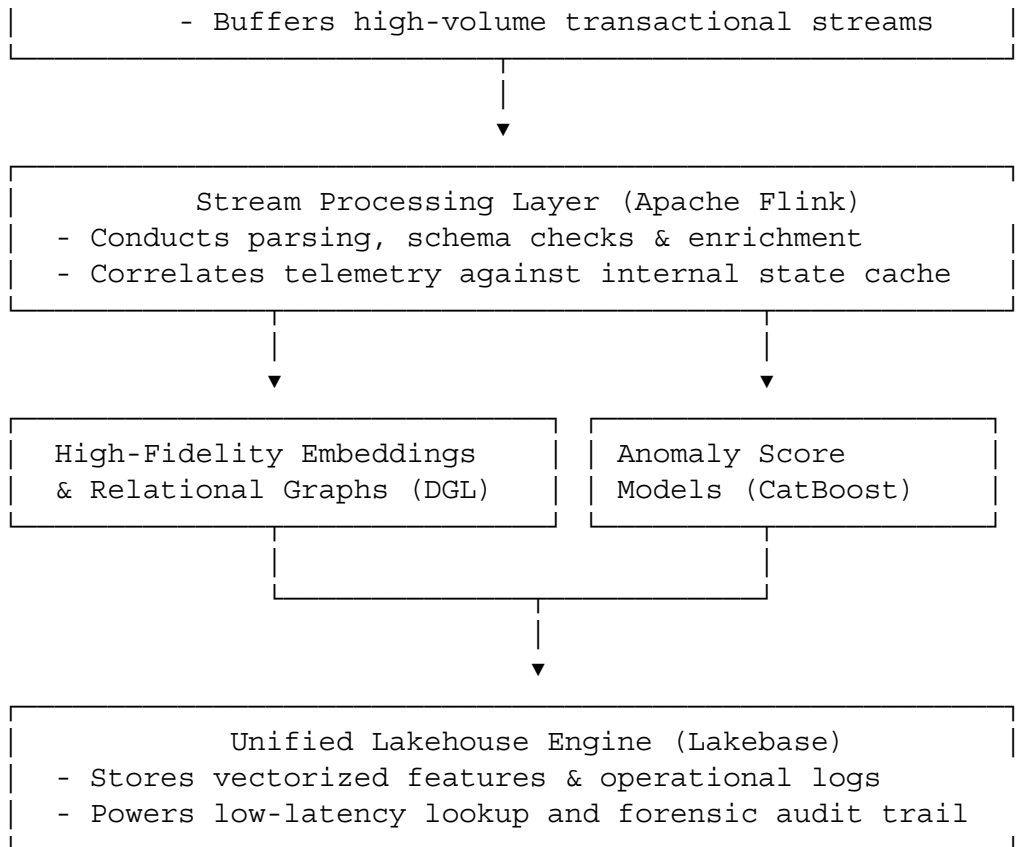
This correlation framework is highly applicable to detecting insider threats and privileged access misuse. In major enterprise resource planning (ERP) systems such as Oracle Fusion, malicious insiders may attempt to alter supplier bank accounts and immediately route payments through the modified account. Unimodal security logs might capture the database configuration change, and transactional tools might flag the outgoing payment, but without contextual correlation, neither alert independently triggers a block.

By correlating identity changes (e.g., granting a privileged role and removing it within a tight window to bypass change management) with real-time transactional requests, the system can dynamically flag unmanaged elevations and prevent internal fraud before funds exit the payment rail.

High-Velocity Stream Processing and Structured Data Standards

The transition to real-time payment rails is accelerated by the adoption of the ISO 20022 messaging standard. Unlike legacy SWIFT MT formats, ISO 20022 features a rich, structured, XML- and JSON-based MX messaging language that allows banks to carry comprehensive parameters, including unambiguous Ultimate Creditor/Debtor details, Legal Entity Identifiers (LEIs), structured invoice references, and specific payment purpose codes. This structured richness improves compliance matching and straight-through processing (STP). However, its implementation introduces the ISO 20022 Paradox. While the richer schemas facilitate precise machine learning checks, the increased complexity and data volumes strain legacy infrastructure and expose translation vulnerabilities during the co-existence era.





Adversaries exploit translation gaps by corrupting fields during conversions from legacy SWIFT MT back into structured MX formats. To secure these transition states, banks utilize stream-processing architectures built on Apache Kafka for real-time queuing and Apache Flink for stateful, in-flight validation. Flink parses the XML messages in flight, checking structural schemas, purpose codes, and routing codes before core posting engines execute the transfers. These real-time transactions are then matched against client identities using secure, zero-data-movement clean rooms. Under platforms like Databricks Lakebase and Snowflake, first-party customer records are resolved against partner identity graphs inside hardware-secured enclaves running Unity Catalog. This integration allows for precise contextual lookup and prevents data exfiltration by keeping raw identifiers hidden throughout the query lifecycle.

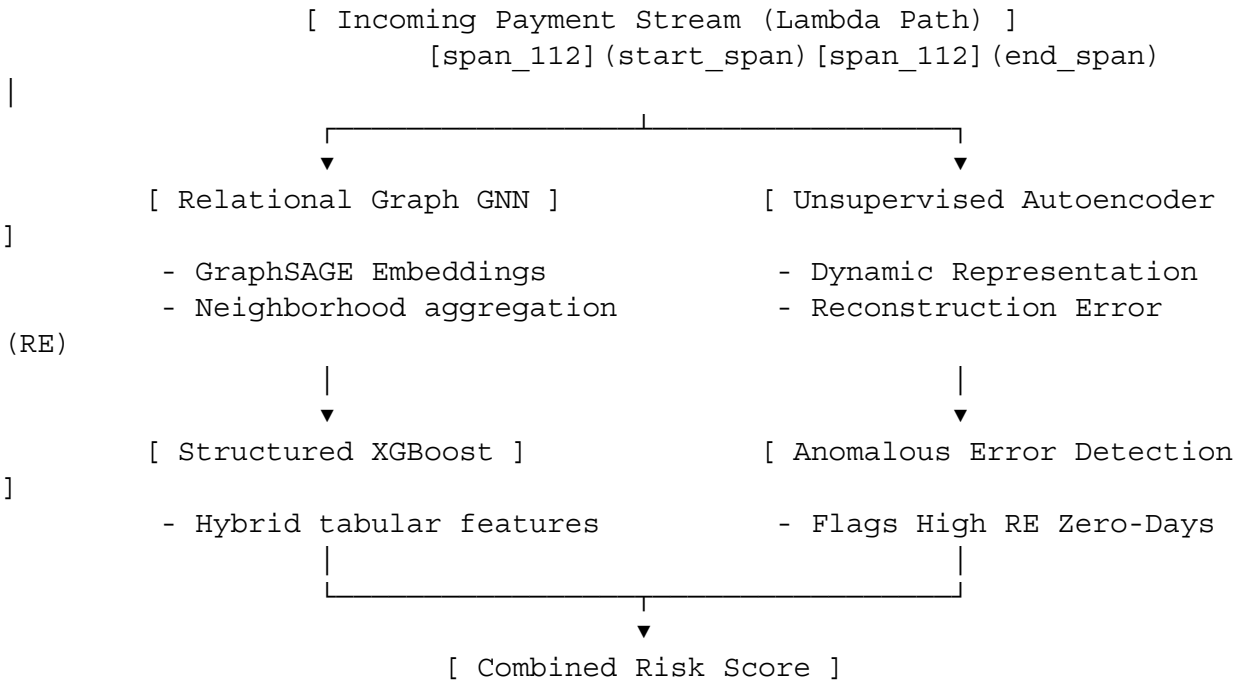
These transactional data platforms use a Postgres-native transactional and vector storage architecture. Search queries are treated as operational workloads; the database hosts a localized NVMe cache for the active vector working sets while keeping older data stored cost-effectively in cold object storage. This system allows security agents to perform hybrid SQL-vector queries directly against payment histories, device logs, and system access lists without executing slow, multi-system ETL processes.

Hybrid Deep Learning and Advanced Anomaly Modeling

Conventional tabular classifiers, such as Logistic Regression, Random Forest, and Gradient

Boosting (including CatBoost and XGBoost), are highly effective at analyzing transactional parameters. These classifiers handle highly imbalanced payment datasets where fraudulent records represent a tiny fraction (often below 0.1%) of total volume. To balance standard mathematical learning, the system normalizes transaction amounts using log transformations and removes redundant variables using Pearson correlation analysis. However, tabular models cannot process network topologies, and they overlook complex relationships like multi-party money-laundering chains or mule networks. To solve this, banks deploy a hybrid architecture combining Graph Neural Networks (GNNs) with Autoencoders. For real-time streaming payments, the framework adopts a Lambda architecture. The system processes incoming data through two parallel paths:

- **The Relational Flow (GNN):** This flow models the digital payment ecosystem as a heterogeneous graph of accounts, merchants, devices, and IP addresses. PyTorch Geometric or Deep Graph Library frameworks execute mini-batch training with neighbor sampling to generate structural embeddings of nodes and edges. Message-passing algorithms aggregate features across a two-hop network neighborhood, which JPMorgan research confirms captures the majority of fraud-related signals. These structural embeddings are fed directly into an XGBoost model, enhancing tabular classification with relational context.
- **The Unsupervised Flow (Autoencoder):** Simultaneously, an Autoencoder processes transaction records. Rather than predicting labels, it is trained exclusively on normal historical transactions to learn a compact representation of standard customer behavior. When an anomalous transaction arrives, its reconstruction error is calculated. High reconstruction errors flag previously unseen fraud patterns, zero-day attacks, or novel money laundering techniques that supervised classifiers would miss.



To coordinate threat intelligence across independent financial institutions without violating data privacy regulations such as GDPR or CCPA, the GNN framework is combined with Federated Learning (FL). Local banks build proprietary transaction graphs and train GNN models locally on their own infrastructure.

Once local iterations are complete, the model parameters (not the customer data) are transmitted to a central coordinator using secure aggregation protocols.

$$\theta_k = \sum_{k=1}^K w_k \theta_k$$

To protect against reverse-engineering or parameter leakage, the central server applies staleness-aware Federated Averaging (FedAvg), where client updates older than two rounds are automatically down-weighted. The framework applies differential privacy (DP) with a Gaussian noise scale ($\sigma = 1.3$), an epsilon budget ($\epsilon = 1.5$), and a delta parameter ($\delta = 10^{-5}$). The global model parameters are updated and redistributed, allowing banks to detect coordinated, cross-institution fraud networks while maintaining data sovereignty and compliance.

| Model Core Type | Algorithmic Strategy | Primary Processing Focus |
 Mathematical/Operational Parameters | False Positive Reduction Impact | :--- | :--- | :--- | :--- |
 :--- || **XGBoost Classifier** | Supervised Ensemble Boosting | Tabular transaction properties and engineered features | Max depth: 9; handles extremely imbalanced datasets via SMOTE. | High for known, recurring fraud patterns. | | **Graph Neural Network (GraphSAGE)** | Relational Neighborhood Aggregation | Multi-relational payment graph topology and edge paths | 2-hop aggregation; mini-batch neighbor sampling. | High; recognizes complex fraud rings and shell companies. | | **Deep Autoencoder** | Unsupervised Reconstruction Anomaly | Re-representation of normal behavioral parameters | Reconstruction error thresholds; trained on normal profiles. | High for zero-day attacks and novel fraud strategies. | | **Federated Learning (FedAvg)** | Privacy-Preserving Collaborative Training | Parameter aggregation across decentralized databases | Differential privacy (ϵ [span_141](start_span)[span_141](end_span)=1.5, δ =10⁻⁵, σ =1.3). | High; prevents cross-border fraud using shared intelligence. |

The Temporal Post-Quantum Threat and Passive Data Staging

Quantum Risks and the Mechanics of Passive Harvesting

The security of global banking relies on asymmetric public-key cryptosystems, such as RSA, Diffie-Hellman, and ECC, to protect TLS sessions, digital signatures, and administrative access. A cryptographically relevant quantum computer (CRQC) running Shor's algorithm can solve prime factorization and discrete logarithms in polynomial time, making these ciphers vulnerable to decryption.

This risk is active through the "Harvest-Now, Decrypt-Later" (HNDL) threat model. Adversaries passively record encrypted transit traffic at internet exchange points or compromised routers. Because this collection requires no active network intrusion or system breach, it leaves no alerts or traditional host-level signatures. The encrypted sessions are archived in bulk storage and will be decrypted retroactively once a CRQC is developed.

The urgency of this threat is analyzed using Mosca's Inequality, which evaluates if the required secrecy duration (X) plus the organizational migration timeline (Y) exceeds the expected time to a CRQC arrival (Z):

$$X + Y > Z$$

Given that migration timelines for complex global banks average 5 to 15 years, and estimates for a CRQC are near the early 2030s, many financial institutions are already in a state of

retroactive compromise.

Data Harvesting, Staging, and Exfiltration Detection

While passive harvesting beyond the bank's network perimeter is difficult to detect, active staging or harvesting of misclassified datasets *within* the bank's internal network generates detectable anomalous patterns. Security teams utilize Microsoft Sentinel and advanced SIEM rules to monitor for internal data-gathering campaigns.

These indicators include:

- **Outbound Callback Connections:** Attackers exploiting vulnerabilities (such as the unauthenticated Langflow API compromise CVE-2026-33017) execute system fingerprinting scripts, base64-encode exfiltrated environment variables, and establish connections to external callback servers or known command-and-control (C2) domains.
- **Out-of-Band Exfiltration:** In advanced SQL injection attacks (such as the MOVEit CVE-2023-34362 exploit), attackers pivot from database discovery to out-of-band exfiltration using custom DNS requests or stacked database functions.
- **The "Smash and Grab" Pattern:** Malicious actors or compromised accounts exfiltrate files by downloading over 200 documents per hour or initiating bulk exports from sensitive SharePoint and OneDrive libraries outside of standard business hours.
- **Privilege Abuse and External Link Exposure:** Compromised accounts frequently attempt to grant high-level API permissions (such as Sites.Read.All or Files.ReadWrite.All) to unauthorized software tools. This is often paired with creating anonymous or public sharing links, allowing data to be exfiltrated directly over the web and bypassing traditional security perimeters.

The SOC Quantum Playbook Use Cases

The Security Operations Center (SOC) Quantum Playbook details five specific operational use cases and corresponding SIEM correlation rules designed to monitor the transition to post-quantum cryptography:

Use Case ID & Focus Area	Target Risk Scenario	Trigger Mechanics & Logic	Priority Score
Use Case 1: Hybrid Downgrade Detection	Adversaries block PQC negotiation, forcing endpoints to fall back to legacy, quantum-vulnerable schemes.	Alerts when a TLS 1.3 handshake between two hybrid-enabled endpoints in the Cryptographic Posture Registry completes using a classical-only cipher suite (e.g., ECDH) without ML-KEM.	High
Use Case 2: Cryptographic Drift Monitoring	Legacy, unapproved, or un-inventoried ciphers are introduced into the production environment.	Scans configurations and TLS handshakes to detect active public-key schemes that do not align with authorized	Medium

Use Case ID & Focus Area	Target Risk Scenario	Trigger Mechanics & Logic	Priority Score
		standards.	
Use Case 3: Certificate Lifecycle Anomalies	Expired, self-signed, or unauthorized certificates bypass automated lifecycle controls.	Detects certificates that fail automated renewal flows, use invalid extensions, or feature non-standard key-usage parameters.	High
Use Case 4: TNFL & Signature Integrity	Trust Anchor Store tampering or compromise of code-signing key repositories.	Monitors modifications to trusted root stores and authenticates the integrity of code-signing signatures across the pipeline.	Critical
Use Case 5: Enhanced HNDL Indicators	Staging, exfiltration, or bulk copying of high-value, long-lived data assets.	Flags bulk archival storage reads, unexpected data exports, or sustained egress from network segments holding data with >10-year secrecy windows.	Critical

A major challenge during hybrid deployment is packet fragmentation. Hybrid ML-KEM and classical ECDH key exchanges require significantly larger public keys and ciphertexts than classical-only handshakes. Because these larger messages exceed standard packet sizes, they frequently get fragmented.

Legacy firewalls and secure web gateways may drop these fragmented packets, leading to handshake failures that mimic downgrade attacks. To avoid false positives, SOC teams must correlate drop errors with packet fragmentation logs.

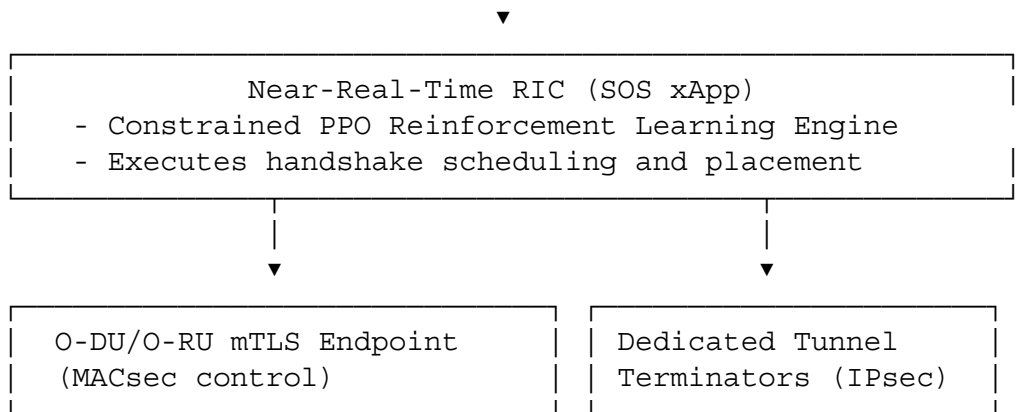
Post-Quantum Optimization and Energy-Aware Orchestration

Migrating to post-quantum cryptography is resource-intensive. The computationally heavy lattice operations of algorithms like ML-KEM (FIPS 203) bottleneck the control plane of high-velocity systems, such as O-RAN (Open Radio Access Network) 5G/6G connections and mobile banking portals, increasing network latency and operational energy consumption.

To manage this, banks implement an energy-aware security orchestration framework within their network operations.

Non-Real-Time RIC (Crypto Policy rApp)
- Defines strategic policy envelope
- Sets allowed PQC suites and rekey bounds

|



The system splits cryptographic management into two functional planes:

1. **Strategic Management:** A Crypto Policy rApp running in the Non-Real-Time RAN Intelligent Controller (Non-RT RIC) defines the high-level policy envelope, specifying the permitted cryptographic suites and maximum rekey intervals.
2. **Tactical Execution:** A Security Operations Scheduling (SOS) xApp running in the Near-Real-Time RIC (Near-RT RIC) translates these rules into tactical schedules. The SOS xApp schedules handshakes to delay non-urgent rekeys into low-load processing windows, manages session resumption, and selects accelerators based on predicted energy efficiency.

To minimize power consumption without degrading security, the SOS xApp uses a constrained Proximal Policy Optimization (PPO) reinforcement learning engine. The environment's reward function is designed to minimize overall power consumption while enforcing safety constraints, such as a maximum 95th percentile (p95) handshake latency of 150\text{ ms} and maintaining the target security level.

This optimization uses a distilled LLM and classifier pipeline within a Generative-Intelligence Cybersecurity Mesh (GI-CSM):

- **Distilled Language Model (Mistral-7B-Pay):** A distilled Mistral-7B base model using Low-Rank Adaptation (LoRA rank = 8, alpha = 16) trained on an 18 GB transaction corpus to parse semantic context.
- **CatBoost Classifier:** Evaluates 64 engineered features and 256 embedding features with a maximum tree depth of 9.
- **PPO Actor-Critic Network:** A 3-layer Multi-Layer Perceptron (MLP) architecture with layer dimensions of 256-128-64, a PPO clipping parameter of 0.2, a value loss coefficient of 0.5, and an entropy bonus of 0.01.

By implementing this framework, the SOS configuration significantly reduces tail latencies compared to baseline operations. Over 60\% of incoming requests resolve as resumed handshakes, decreasing computing overhead. This architecture keeps operations within performance SLAs while maintaining a zero-trust model where cryptographic enforcement remains at standards-compliant endpoints.

Explainable AI and Cross-Regulatory Alignment

Algorithmic Explanations and Human-in-the-Loop Integration

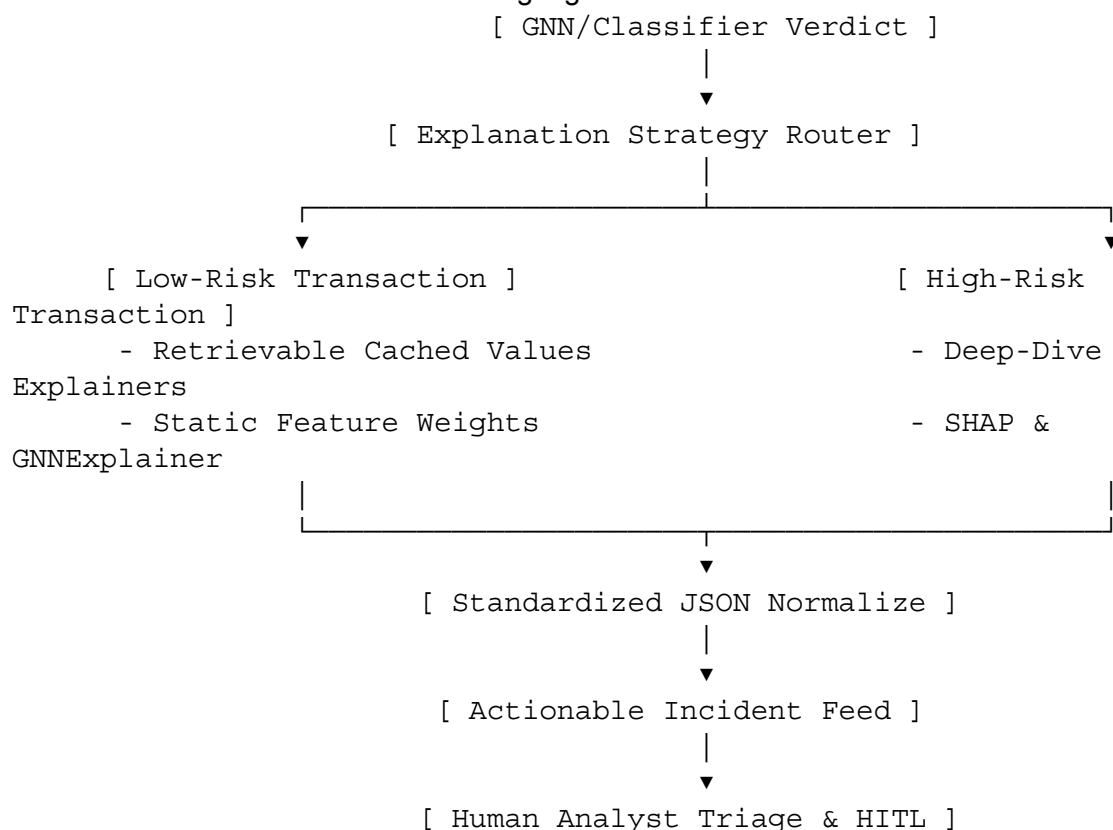
As financial institutions transition to automated, AI-driven transaction monitoring and threat classification, they must maintain model explainability and transparency. Regulated banking environments require that automated decisions (such as transaction blocks or security alerts) can be clearly justified to compliance officers, auditors, and customers. To achieve this, models are paired with Explainable AI (XAI) frameworks.

The explainability pipeline maps high-dimensional machine learning outputs into three structured layers of translation:

- **Global Interpretability (SHAP):** SHAP assigns contribution scores based on cooperative game theory to establish a stable, global model baseline. This allows model risk management teams to audit feature importance and identify potential biases before production deployment.
- **Local Interpretability (LIME & EBMs):** LIME approximates the local decision boundary of a target transaction by fitting a simpler, interpretable linear surrogate model. This generates fast, real-time explanation scores for SOC triage dashboards.
- **Structural and Subgraph Interpretability (GNNExplainer):** For relational models, GNNExplainer isolates the minimal subgraph and edge mask that drove the classification. It translates these graph-based outputs into plain-language business narratives.

These explanations are normalized into a standard JSON format and routed through an Explanation Strategy Router. The system balances computing resources by tailoring the depth of the explanation to the risk level of the transaction. Low-risk alerts retrieve cached feature metrics, while high-value transactions flagged for manual review are routed to high-fidelity explainers for complete forensic analysis.

This output is integrated into a continuous Human-in-the-Loop (HITL) learning system. When a human investigator confirms an alert or overrules a decision, the feedback loop updates the detection models and refines the routing logic.



|
▼
[Retraining Parameter Feed]

Strict Governance, Hard Limits, and Incident Playbooks

To safeguard automated agent interactions and prevent unmanaged actions, banks enforce operational controls:

- **Operational Limits & Controls:** The platform enforces rate limits and transaction thresholds. If an AI agent detects a transactional or security anomaly, the system executes an automated pause. In critical scenarios, administrators can activate a break-glass disable path to suspend autonomous operations.
- **Multi-Agent Trust Boundaries:** Each AI agent is assigned a unique, role-based security profile, preventing implicit privilege escalation during agent-to-agent data transfers.
- **On-Chain Verification:** To ensure integrity and non-repudiation, model checkpoints and system logs are hashed (using SHA-256) and recorded on a permissioned blockchain (ModelLedger). Transactions must be validated with post-quantum CRYSTALS-Dilithium signatures.
- **Incident Response Integration:** The SOC maintains four pre-configured incident response playbooks to address post-quantum transition scenarios:
 1. *PQC Algorithm Vulnerability Disclosure:* Activates agile algorithm rotation policies if a deployed post-quantum scheme is broken or degraded.
 2. *Confirmed Hybrid Downgrade Attack:* Isolates compromised endpoints and establishes secure, verified mTLS channels when a downgrade attempt is detected.
 3. *Credible CRQC Announcement:* Speeds up the migration roadmap for critical data assets with long-term secrecy requirements.
 4. *Emergency Algorithm Rotation:* Executes rapid, automated key and certificate rotation across all production-level enclaves.

Future-Ready Analytical Architecture and Recommendations

To secure the digital banking enterprise against emerging transactional, identity, and post-quantum threats, financial institutions should implement the following structural and technical recommendations:

1. Unified Multimodal Ingestion Pipeline

To eliminate traditional security silos, organizations must integrate all risk domains into a shared analytical environment. The ingestion architecture should utilize high-throughput Apache Kafka queues and stateful Apache Flink engines to standardize and parse structured ISO 20022 payment streams in flight. This pipeline must continuously correlate cybersecurity alerts (MFA session cookie theft, active directory updates, and unusual API activities) with active payment transactions and behavioral biometric telemetry.

2. Implement Hybrid GNN-Autoencoder Real-Time Models Banks should deploy a dual-track analytical modeling framework:

- A relational Graph Neural Network (GNN) to construct real-time account and merchant embeddings, mapping multi-hop interactions to isolate fraud rings and money-laundering subgraphs.
- An unsupervised Autoencoder to compute reconstruction errors on all streaming transactions, flagging zero-day attacks and novel fraud strategies without relying on pre-labeled data.

3. Establish Privacy-Preserving Federated Model Collaboration

To defend against cross-border fraud and structured money-laundering rings, financial institutions should coordinate detection modeling via secure Federated Learning. Utilizing decentralized local payment graphs and secure aggregation protocols (complying with GDPR and CCPA) allows institutions to aggregate updates using staleness-aware FedAvg algorithms and robust differential privacy parameters.

4. Execute a Structured Post-Quantum Cryptographic Migration

Banks must immediately prioritize post-quantum migration by separating execution into parallel tracks:

- *Track A (Immediate Deployment)*: Secure data-in-transit pathways carrying long-lived sensitive records using hybrid ML-KEM to defend against passive Harvest-Now, Decrypt-Later threats.
- *Track B (Strategic Transition)*: Build a comprehensive, centralized Cryptographic Posture Registry and document all certificate authorities, key stores, and algorithm dependencies. Establish cryptographic agility as a permanent operational capability, enabling the automated rotation of algorithms without requiring major code changes.

5. Deploy SOC Quantum Playbooks & Validate Infrastructure Performance

The Security Operations Center must develop and operationalize the five core SOC quantum use cases, with a primary focus on detecting hybrid downgrade attacks and anomalous bulk data exfiltration. Network engineers should run comprehensive tabletop exercises to validate that legacy middleboxes, gateways, and load balancers can process fragmented, post-quantum handshake packets without causing operational downtime or false-positive fallback events.

6. Embed Contextual and Regulatory-Grade Explainable AI

To satisfy global compliance frameworks (such as DORA, the EU AI Act, and GDPR Article 22), automated decision engines must be integrated with post-hoc explainers. Utilizing SHAP for global model validation, LIME for local transaction-level triage, and GNNExplainer to map relational subgraphs allows banks to present clear, plain-language justifications to risk officers, auditors, and consumers. This framework should be anchored by a continuous Human-in-the-Loop learning loop to continuously refine detection models and strategy routers based on analyst feedback.

Works cited

1. Data, AI, and trust: The next era of financial crime prevention | Asian Banking & Finance, <https://asianbankingandfinance.net/financial-technology/commentary/data-ai-and-trust-next-era-financial-crime-prevention> 2. Cybersecurity Solutions | SIEM, SOC AI, GRC & Threat Intelligence - CyberSilo, <https://cybersilo.tech/solutions> 3. What Is an SQL Injection? - Palo Alto Networks, <https://www.paloaltonetworks.com/cyberpedia/sql-injection> 4. The silent heist: cybercriminals use information stealer malware to compromise corporate networks | Cyber.gov.au, <https://www.cyber.gov.au/about-us/view-all-content/alerts-and-advisories/silent-heist-cybercriminals-use-information-stealer-malware-compromise-corporate-networks> 5. Multimodal Fraud Detection Systems: Integrating Transactional, Behavioral, and Cybersecurity Data for Enhanced Threat Intelligence - ResearchGate, https://www.researchgate.net/publication/403159027_Multimodal_Fraud_Detection_Systems_Integrating_Transactional_Behavioral_and_Cybersecurity_Data_for_Enhanced_Threat_Intelligence 6. Investigating Document Exfiltration with the Graph Activity Log | Practical365, <https://practical365.com/investigating-document-exfiltration-with-the-graph-activity-log/> 7. Bank of Maharashtra Launches FinSpark'26, a National-Level Cybersecurity Hackathon to Secure the Future of Banking - NewsVoir, <https://www.newsvoir.com/release/bank-of-maharashtra-launches-finspark-26-a-national-level-cybersecurity-hackathon-to-secure-the-future-of-banking-36151.html> 8. Bank of Maharashtra Launches FinSpark'26, a National-Level Cybersecurity Hackathon to Secure the Future of Banking, <https://www.hinditvnews.co.in/bank-of-maharashtra-launches-finspark26-a-national-level-cybersecurity-hackathon-to-secure-the-future-of-banking/> 9. Monitoring and Incident Response for Oracle Fusion Cloud (ERP and HCM): Audit Trails, Alerting, and SIEM Integration | by Khader Nawaz khan | Medium, <https://medium.com/@khadernawazkhan/monitoring-and-incident-response-for-oracle-fusion-cloud-erp-and-hcm-audit-trails-alerting-and-dd5a1f2d89a3> 10. AI-Powered Cybersecurity Mesh for Financial Transactions: A Generative-Intelligence Paradigm for Payment Security - MDPI, <https://www.mdpi.com/2813-0324/12/1/10> 11. ISO 20022: Security and Compliance in the New Global Messaging Standard - Medium, <https://medium.com/@viktor.davydych/iso-20022-security-and-compliance-in-the-new-global-messaging-standard-63450ee23cb5> 12. Best Practices for MFT-Enabled Batch Settlements in Banking - Kiteworks, <https://www.kiteworks.com/managed-file-transfer/mft-batch-settlements-banking-practices/> 13. BFSI GCC Talent Report - Quess Corp, <https://www.quescorp.com/reports-pdf/BFSI-GCC-Report.pdf> 14. Announcing New OpenSharing and Marketplace capabilities for the AI era | Databricks Blog, <https://www.databricks.com/blog/announcing-new-opensharing-and-marketplace-capabilities-ai-era> 15. Introducing the Agentic CDP: A New Species of CDP for a New Era of Agents - Databricks, <https://www.databricks.com/blog/what-is-an-agentic-cdp> 16. Announcing Lakebase Search: agent-native retrieval built into Lakebase Postgres | Databricks Blog, <https://www.databricks.com/blog/announcing-lakebase-search-agent-native-retrieval-built-lakebase-postgres> 17. Announcing Databricks Lakebase Launch Partners, <https://www.databricks.com/blog/announcing-databricks-lakebase-launch-partners> 18. (PDF) Intelligent Fraud Detection: Applying Advanced Analytics and Cybersecurity Insights in U.S. Finance - ResearchGate,

https://www.researchgate.net/publication/397530946_Intelligent_Fraud_Detection_Applying_Advanced_Analytics_and_Cybersecurity_Insights_in_US_Finance 19. Optimizing Fraud Detection in Financial Services with Graph Neural Networks and NVIDIA GPUs, <https://developer.nvidia.com/blog/optimizing-fraud-detection-in-financial-services-with-graph-neural-networks-and-nvidia-gpus/> 20. Supercharging Fraud Detection in Financial Services with Graph Neural Networks (Updated) | NVIDIA Technical Blog, <https://developer.nvidia.com/blog/supercharging-fraud-detection-in-financial-services-with-graph-neural-networks/> 21. Enhancing Fraud Detection in Banking with Deep Learning Graph Neural Networks and Auto Encoders for Real-Time Credit Fraud Prevention - Science Excel, <https://www.sciencexcel.com/articles/XHULxq1J4yLokpbwsKtsZhCfDOjenR3mrOpRit.pdf> 22. A Hybrid Graph Neural Network and Explainable AI Framework for Financial Fraud Detection, <https://cspub-ijcisim.org/index.php/ijcisim/article/view/2877> 23. Graph Neural Network-Based Framework for Real-Time Financial Fraud Detection in Digital Payment Ecosystems - ResearchGate, https://www.researchgate.net/publication/400595442_Graph_Neural_Network-Based_Framework_for_Real-Time_Financial_Fraud_Detection_in_Digital_Payment_Ecosystems 24. Fraud detection with GNN - Kaggle, <https://www.kaggle.com/code/jawherjabri/fraud-detection-with-gnn> 25. Graph Neural Networks for Fraud Detection: How to Guide - FluxForce AI, <https://www.fluxforce.ai/blog/graph-neural-networks-for-fraud-detection> 26. Interpretable Ensemble Learning Models for Credit Card Fraud Detection - Preprints.org, <https://www.preprints.org/manuscript/202509.1170/v1/download> 27. Real-Time Fraud Detection Using Graph Neural Networks and Federated Learning, <https://ijeret.org/index.php/ijeret/article/view/304> 28. Explainable Graph-Based Financial Fraud Detection Using Machine Learning and Data Analytics - IRJIET, <https://irjiet.com/article/Explainable-Graph-Based-Financial-Fraud-Detection-Using-Machine-Learning-and-Data-Analytics/3197> 29. Harvest now, decrypt later: how the collection actually works - Hive Storm, <https://hivestorm.io/insights/how-harvest-now-decrypt-later-actually-works> 30. A Banker's Guide to Quantum Safe Cryptography - Part 2: A Banker's Guide to Quantum Safe Cryptography: Roadblocks to PQC Migration and Strategic Solutions for Financial Institutions | Cryptomathic, <https://www.cryptomathic.com/blog/a-bankers-guide-to-quantum-safe-cryptography-part-2> 31. Incident Response Readiness in the Era of Quantum-Enabled Cryptographic Failure - International Journal of Science and Research Archive, http://ijsra.net/sites/default/files/fulltext_pdf/IJSRA-2026-1169.pdf 32. Harvest-Now, Decrypt-Later: A Temporal Cybersecurity Risk in the Quantum Transition, <https://www.mdpi.com/2673-4001/6/4/100> 33. A Multi-Plane Detection Framework for Quantum-Enabled Cryptographic Attacks: Operationalizing the Detection of Harvest-Now-Decrypt-Later Activity and Attacks on Post-Quantum Cryptography - ResearchGate, https://www.researchgate.net/publication/408157861_A_Multi-Plane_Detection_Framework_for_Quantum-Enabled_Cryptographic_Attacks_Operationalizing_the_Detection_of_Harvest-Now-Decrypt-Later_Activity_and_Attacks_on_Post-Quantum_Cryptography 34. What Is Harvest Now, Decrypt Later (HNDL)? - Keyfactor, <https://www.keyfactor.com/education-center/what-is-harvest-now-decrypt-later-hndl/> 35. Guide to PQC Migration: Prepare for the Quantum Computing Era | QNu Labs, <https://www.qnulabs.com/blog/guide-to-pqc-post-quantum-cryptography-migration> 36. Why You Need a Cryptographic Inventory Before You Start PQC Migration, <https://quantumsecuritydefence.com/insights/cryptographic-inventory/> 37. Quantum Sundays |62

Enterprise Blueprint for Post-Quantum Cryptography Migration | by Adnan Masood, PhD. | Medium, <https://medium.com/@adnanmasood/quantum-sundays-62-enterprise-blueprint-for-post-quantum-cryptography-migration-3e75e9d85136> 38. Microsoft cloud security benchmark v2 - Data Protection, <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-v2-data-protection> 39. The SOC Quantum Playbook: Detection Rules, Threat Intelligence, and Incident Response, <https://postquantum.com/post-quantum/soc-quantum-security-pqc-operations/> 40. CVE-2026-33017: How attackers compromised Langflow AI pipelines in 20 hours | Sysdig, <https://www.sysdig.com/blog/cve-2026-33017-how-attackers-compromised-langflow-ai-pipelines-in-20-hours> 41. PQC Migration: What You Discover After You Begin - Encryption Consulting, <https://www.encryptionconsulting.com/pqc-migration-what-you-discover-only-after-you-begin/> 42. Solving the Post-Quantum Control Plane Bottleneck: Energy-Aware Cryptographic Scheduling in Open RAN - arXiv, <https://arxiv.org/html/2602.11820v1> 43. What is Explainable AI (XAI)? - IBM, <https://www.ibm.com/think/topics/explainable-ai> 44. What Is Explainable AI (XAI)? | AI Glossary - Lumenova AI, <https://www.lumenova.ai/ai-glossary/explainable-ai/> 45. Interpretable Ensemble Learning Models for Credit Card Fraud Detection - MDPI, <https://www.mdpi.com/2076-3417/15/22/12073> 46. Explainable AI in cybersecurity: How it works and why it matters - N-iX, <https://www.n-ix.com/explainable-ai-in-cybersecurity/> 47. Explainable AI in Big Data Fraud Detection - arXiv, <https://arxiv.org/html/2512.16037v1> 48. Explaining GNN Predictions for Compliance | PyG Guide - Kumo.ai, <https://kumo.ai/pyg/production/explainability/> 49. Explainable Artificial Intelligence in Financial Fraud Detection: A Systematic Review and FinTech-Oriented ADO–TCCM Meta-Framework for Trust, Governance, and Transparency - MDPI, <https://www.mdpi.com/2674-1032/5/3/60> 50. (PDF) Evaluating explain ability frameworks (e.g., SHAP, LIME) in fraud detection models, https://www.researchgate.net/publication/395136073_Evaluating_explain_ability_frameworks_e_g_SHAP_LIME_in_fraud_detection_models 51. Fusion AI Agents: Secure by Operations Best-Practice Considerations | cloud-infrastructure, <https://blogs.oracle.com/cloud-infrastructure/fusion-ai-agents-secure-by-operations> 52. How Blockchain Enhances SIEM for Comprehensive Threat Detection - SearchInform, <https://searchinform.com/articles/cybersecurity/measures/siem/use-cases/blockchain/> 53. How to Start Your Enterprise Post Quantum Cryptography Migration Plan, <https://www.encryptionconsulting.com/how-to-start-your-enterprise-post-quantum-cryptography-migration-plan/> 54. PQC Migration Framework v2.0 Released: What Changed, <https://postquantum.com/security-pqc/pqc-migration-framework-v2/>